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SPORTS PERFORMANCE

The effectiveness of a practical half-time re-warm-up strategy on performance and the physical response to soccer-specific activity

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ABSTRACT

The aim of this study was to assess the influence of a half-time (HT) re-warm up (RWU) strategy on measures of performance and the physical and perceptual response to soccer-specific activity. Ten male soccer players completed a control (CON) and RWU trial, in which participants completed 60 min (4 x 15-min periods with a 15-min HT interspersing the third and fourth periods) of a soccer-specific exercise protocol. The CON trial comprised a passive 15-min HT, whilst the RWU trial comprised a passive 12-min period, followed by a 3-min RWU. The RWU elicited an improvement in 20 m sprint times (d= 0.6; CON: 3.42 ± 0.20 s; RWU: 3.32 ± 0.12 s), and both squat (d= 0.6; CON: 26.96 ± 5.00 cm; RWU: 30.17 ± 5.13 cm) and countermovement jump height (d= 0.7; CON: 28.15 ± 4.72 cm; RWU: 31.53 ± 5.43 cm) following the RWU and during the initial stages of the second half. No significant changes were identified for 5 m or 10 m sprint performance, perceived muscle soreness, or PlayerLoad™. Ratings of perceived exertion were however higher (~2 a.u) following the RWU. These data support the use of a HT RWU intervention to elicit acute changes in performance.

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KEYWORDS
Jump Performance; Intermittent Activity; Sprint Performance; Active Recovery

Introduction

Soccer matches are commonly divided into two 45-min halves, separated by a 15-min half-time (HT) period. Typically, this HT period is reserved for recovery, treatment, and tactical discussion, and is primarily passive in nature (Russell, West, Harper, Cook, & Kilduff, 2015b). However, research into elite-level match-play has shown that key performance markers are reduced during the first fifteen minutes of the second half, when compared to the first 15 min of the first half (Bradley et al., 2009; Di; DI Salvo, Gregson, Atkinson, & Drust, 2009; Lovell, Barrett, Portas, & Weston, 2013). Consequently, it has been suggested that a purely passive HT period may not prove to be beneficial to performance (Edholm, Krustrup, & Randers, 2015; Mugglestone, Morris, Saunders, & Sunderland, 2013; Russell, Tucker, Cook, Giroud, & Kilduff, 2018; Russell et al., 2015a). To overcome this issue, the implementation of HT re-warm-up strategies (RWU) have been proposed to offer a method of better preparing the players for the second half of match-play (Russell et al., 2015b).

It has been suggested that although 89% of practitioners acknowledge the potential benefits of a HT RWU strategy, only 58% actually instruct their players to complete one (Towlson, Midgley, & Lovell, 2013). This disparity has been attributed to pressured time constraints and reluctance from the head coach/manager to willingly reduce time dedicated to tactical interventions (Towlson et al., 2013). Where studies have previously assessed the effectiveness of a HT RWU strategy, there has been no clear consensus on the most beneficial and practical methods. For example, previous studies have identified that both a 5- (Lovell, Kirke, Siegler, Mcnaughton, & Greig, 2007, Lovell, Midgley, Barrett, Carter, & Small, 2013) and 7 min (Mohr, Krustrup, Nybo, Nielsen, & Bangsbo, 2004) RWU strategy appear to elicit beneficial changes in performance when compared to a passive HT strategy. However, the duration of these RWU methods is far in excess of what has previously been suggested as a realistic time-frame to be able to implement a RWU strategy, with practitioners suggesting the time available would be as little as 3 min (Towlson et al., 2013). Recently, a 3-min cycling-based RWU strategy was also shown to be as effective as a 7-min RWU for improving intermittent sprint performance (Yanaoka et al., 2018).

Previous studies have also used methods to RWU players which may practically be difficult to deliver to a whole squad during this constrained time period. For example, previous studies have utilised whole body vibration methods (Lovell et al., 2007, Lovell, Midgley, et al., 2013), leg press machinery (Zois, Bishop, Fairweather, Ball, & Aughey, 2013), and heated garments (Russell et al., 2018). Although these methods are supported with regards to their potential benefits on subsequent performance, the feasibility of these methods to be applied to a whole team of players is somewhat questionable. It has been suggested that observed benefits from a HT RWU can be attributed to the ability to elicit post-activation potentiation (PAP) within the lower limbs of players (Russell et al., 2015b). Heavy resistance exercise are often utilised to elicit the most beneficial PAP effects; however, as previously suggested, these methods are not always practical for a HT RWU. Alternatively, both ballistic and plyometric exercises have been shown to elicit PAP induced improvements in both sprint and jump performance (Turner, Bellhouse, Kilduff, & Russell, 2015), whilst also offering a method which can be administered

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to a whole team during a reduced time frame. In further support of this, a recent review article by Silva, Neiva, Marques, and Izquierdo (2018) advocates the use of 2–5 min of high intensity (>90% of maximum heart rate) jump and sprint activity as an effective RWU strategy.

Assessing the effectiveness of any within match strategy can present many issues. For example, although the use of actual match-play offers ecological validity, the contextual factors which exist during match play often makes it difficult to attribute changes in physical responses and performance to the application of any intervention (Edholm et al., 2015). It is for this reason that previous studies have often utilised soccer-specific exercise protocols (SSEP) to provide a standardised yet valid response to soccer-specific activity whilst allowing for the assessment of controlled measures (Bendiksen et al., 2012; Nicholas, Nuttall, & Williams, 2000; Small, Mcnaughton, Greig, & Lovell, 2009; Williams, Abt, & Kilding, 2010).

There is, therefore, a need to develop a HT RWU strategy that is both time-efficient and practically applicable to contemporary soccer, and assess the response to the RWU using a standardised SSEP, and contemporary measures. The aim of this study was to assess the performance, physical, and perceptual responses associated with the completion of an ecologically valid 3-min HT RWU strategy when compared to a passive HT period. It was hypothesised that the HT RWU would elicit beneficial responses during the initial stages of the second half of the protocol.

Method

Research design

The experimental research design used in the current study was a within-participants cross-over design to assess changes in physical performance and both the perceptual and physical response to soccer-specific activity. Participants were required to attend for testing on three separate occasions, ensuring that a minimum of 72 h separated each session (Ispirlidis et al., 2008). The three sessions included a familiarisation trial, and two experimental trials comprising 60 min of a SSEP (Lovell, Knapper, & Small, 2008). The experimental trials were completed with a counterbalanced measures design. The standardised activity of the current protocol thus allowed for a direct comparison of the performance, perceptual, and physical response to the completion of a passive HT period (CON trial) and a HT RWU strategy. The dependent variables were chosen to quantify response to the experimental trials by using contemporary measurements which are regularly used in an applied setting (Halson, 2014). For example, perceptual measures and heart rate are often used in applied practice to monitor internal load, with modern developments in microelectrical measurement systems (MEMS) offering methods to quantify external load. It has recently been advocated that MEMS derived accelerometry-based metrics could offer reliable methods (casamichana, Castellano, Calleja-Gonzalez, Roman, & Castagna, 2013; Ehrmann, Duncan, Sindhusake, Franzsen, & Greene, 2016) to assess changes in movement efficiency during soccer-specific activity (Page, Marrin, Brogden, & Greig, 2015, 2016, 2019), thus providing a method to assess the effectiveness of contemporary interventions such as a HT RWU strategy.

Participants

Ten male amateur soccer players (mean ± SD: age 23 ± 4 yrs, height 182.0 ± 6.4 cm, mass 77.3 ± 7.2 kg) participated in the current study. The sample size selected was identified using an *a priori* calculation from pilot study data. This was identified as being sufficient to provide appropriate statistical power (0.8; $p = 0.05$) to evaluate the interactions for all independent variables. In addition to weekly matches, participants completed training volumes of ~3-4 hours·week^{-1}. Participants were apparently healthy and were injury-free for a minimum of 3 months prior to testing. All participants provided informed consent before commencing any trials.

A health screening procedure was completed prior to each trial, with this comprising a health and well-being questionnaire and the measurement of resting heart rate (HR) and blood pressure (BP) (Omron, MX3 Plus, Netherlands). Resting values of >90 beats·min^{-1} and >140/90 mm/hg for HR and BP, respectively, were contraindications to exercise. The current study was ethically approved by an institutional ethics committee.

Experimental procedures

All participants were required to complete three separate trials comprising one familiarisation trial, and two separate experimental trials. To remove surface interactions, all trials were conducted on the same indoor playing surface. All testing was completed between 12 and 5 pm to mimic regular match-day routines, and to regulate any circadian rhythm variations (Teo, Newton, & Mcguigan, 2011). Participants were asked to repeat similar nutritional intake, before the second experimental trial, of that which was consumed in the preceding 48 h of the first trial. Participants were instructed to abstain from the consumption of alcohol (Barnes, 2014), and from performing the exhaustive exercise in the preceding 72 h before a trial.

The familiarisation trial consisted of a 15-min soccer-specific warm-up routine using the Raise, Activate, Mobilise, and Potentiate (RAMP) method (Jeffreys, 2007), and the completion of a 30-min bout of a SSEP (Lovell et al., 2007). During the familiarisation trial, the participants were also instructed on the correct use of all testing equipment.

The two experimental trials comprised a CON trial and a RWU trial, the order of which was randomised. Before each trial, a 15-min RAMP warm-up was conducted, to prepare for the following exercise bout. In an attempt to best replicate typical pre-match practices, a 15-minute period interspersed the completion of the warm-up period and the start of each trial. During the CON trial, participants completed the first 45 min of the SSEP, followed by a passive 15-min HT interval. During the HT period participants were instructed to remain seated and consume water *ad libitum*. Total fluid consumption in the first experimental trial was recorded (403 ± 198 ml), and was replicated in the subsequent trial. The RWU trial comprised the same warm-up and first half routine; however, the initial 12 min of the HT interval were passive, followed by a 3-min re-warm-up strategy.

The RWU protocol included a combination of bodyweight exercises, as well as ballistic and plyometric movements (Russell et al., 2018). The RWU protocol was performed over a 20 m circuit utilising a RAMP method (Jeffreys, 2007). The participants were instructed to perform two sets of jogging (40 m) and skipping (40 m), followed by one set (20 m) of each dynamic stretch exercise (knee raises; heel flicks; lateral side lunges; front lunges; high kicks; tuck jumps; and reactive sprints). The movements were performed in tandem with an example being provided by the lead researcher. Following the completion of the aforementioned dynamic stretches, participants completed two sets each of straight sprinting, tuck jump into sprints and reaction sprints (facing opposite way and then turning and sprinting). For both experimental trials, a further 15 min of the soccer simulation was then completed following the HT interval. Figure 1 provides a schematic representation of the experimental design and measures.

Soccer simulation

The SSEP utilised in the current study was the Soccer Aerobic Field Test (SAFT^{90}) (Lovell et al., 2008). The protocol comprised a 20-m agility-based course with slalom training poles positioned at the 2-m, 9-m, 10-m, and 11-m marks. Participants were required to either run backwards or side-step around the 2-m pole, before running forwards and slaloming between the 9-m, 10-m, and 11-m poles at varied intensities. When the participants reached the 20-m marker they were instructed to return to the start at the set intensity whilst bypassing the poles. The locomotive activity associated with the protocol is based on time-motion analysis data obtained from professional match-play, and is controlled through verbal cues on an audio CD. The entire protocol consists of a 15-min activity profile which can be repeated six times to create a full 90-min simulation. The current study only repeated the activity profile four times, as performance measures were only assessed in the opening 15 min of the first and second half. Comparisons between these two periods can indicate decrements in performance following the HT period (Lovell et al., 2007; Mohr et al., 2004).

Experimental measures

A tri-axial accelerometer (Kionix KX94; Kionix, Ithaca, NY, USA) recording at 100 Hz was contained within an inertial measurement unit (MinimaxX, S4; Catapult Innovations, Scoresby, Australia), and was housed within a standardised neoprene pouch at the cervical region of the participants spine. Triaxial accelerometry data were continuously recorded during each experimental trial to quantify tri-axial (PL_{total}) and uniaxial PlayerLoad™ in the medial-lateral (PL_{ML}), anterior-posterior (PL_{AP}), and vertical (PL_V) movement planes. All PlayerLoad™ measures were calculated using Catapult Sprint software v5.0.9.2 (Catapult Innovations) for the 0–5, 5–10, 10–15, 45–50, 50–55, and 55-60-minute periods of the experimental trials (Figure 1). The PlayerLoad™ metrics and associated calculations have previously been defined in a number of research studies (Boyd, Ball, & Aughey, 2011; Page et al., 2015, 2016, 2019). Heart rate values were recorded as point readings immediately pre-trial (0 min), every 5 min in the first 15-min period of the first half, immediately pre HT, immediately pre second half, and every 5 min in the first 15-min period of the second half.

Three single beam timing gates (Smartspeed, Fusion Sport, Australia), set at a standing torso height were used to assess the speed of 5 m, 10 m, and 20 m sprints. Participants were required to perform three maximal sprints from a standing position with the front foot placed in line with the first timing gate (Ishøi et al., 2017). Timing started when the participants passed the first timing gate at 0-m and was recorded when they passed each of the subsequent gates. Only the best attempt (least time taken to complete the required distance) was considered for analysis. As illustrated in Figure 1, sprint performance was recorded at rest, at 5-min sections throughout the first 15-min period of the SSEP, immediately pre- and post HT, and for each 5-min section of the first 15 min of the second half.

Immediately following each of the aforementioned sprint performance assessments, the participants were also required to complete 6 maximal effort jumps (3 x SJ and 3 x CMJ) using a jump mat (Smart Jump, Fusion Sport, Australia). All jumps were standardised by giving participants the same verbal instructions. For the SJ, the participants were initially instructed to move into a squat position with a 90-degree bend at the knee. They were then told to hold this position for a 2-s period, before then performing a maximal vertical jump. The same instructions were provided for the CMJ; however, for these jumps, the participants were told to not hold the squat position and instead instantly initiate the vertical jump. For both types of jumps, the participants were instructed to keep their hands on their hips,

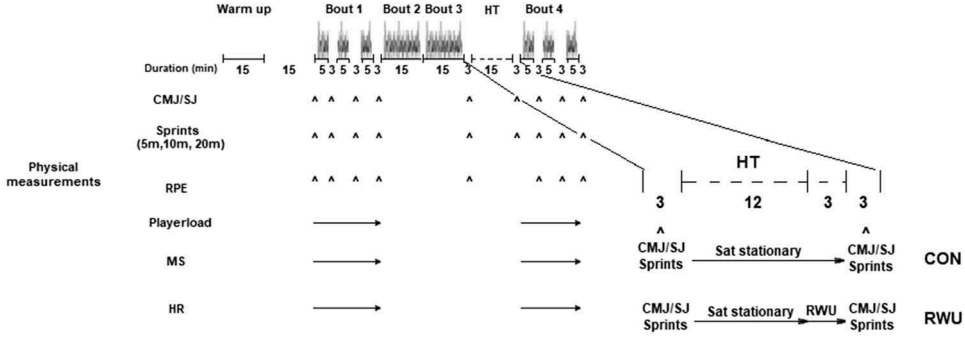

Figure 1. Schematic representation of the experimental design, durations, and measures. Vertical arrows depict point measurements, and horizontal arrows depict values recorded continuously. The zoomed image depicts a representation of the HT procedures.

and once airborne keep their legs straight until they contact the ground (Arnason et al., 2004; Stølen, Chamari, Castagna, & Wisløff, 2005). For each jumping assessment, only the best attempt (highest jump height) was considered for analysis. A 20-s rest period was allocated between each sprint and jump effort (Russell et al., 2018), thus resulting in a 3-min testing period for each performance assessment (illustrated in Figure 1).

As identified in Figure 1, perceptual measures were also recorded as point readings throughout the experimental trials. The perceptual measures comprised the assessment of the participants muscle soreness (MS) and rating of perceived exertion (RPE). The RPE data were recorded using the Borg 6–20 point scale (Borg, 1982) immediately prior to the SSEP, at 5-min increments throughout the first 15-min period, immediately prior to HT, and at each 5-min increment in the first 15-min period of the second half. To quantify lower limb MS, participants were instructed to perform a single squat down to a position of 90-degrees of knee flexion, before using a marker to draw a straight vertical line on a visual analogue scale (VAS) (Chen & Nosaka, 2006). The VAS scale comprised a 0–10 cm horizontal line, with the words "not sore at all" at the 0-cm mark and "very, very sore" at the 10-cm mark. The MS measures were completed at the same time periods as the RPE data, with an additional recording being made immediately prior to the start of the second half.

Statistical analyses

The assumptions of the general linear model (GLM) were initially assessed to ensure model adequacy. A two-way repeated-measures GLM was chosen as an appropriate parametric test to compare differences between trials and measurement points. Where significant main effects and interactions were identified, post hoc pairwise comparisons with a Bonferroni correction were completed. For all significant main effects and interactions, 95% confidence intervals for difference (CI diff) are reported. Partial eta squared values are also reported for all main effects and interactions, with these classified as small (0.01–0.059), moderate (0.06–0.137), and large (>0.138) (Richardson, 2011). Additionally, Cohens d effect sizes and 95% confidence intervals (CI) have been calculated to compare all measures recorded between trials. Cohen's d (*d*) effect sizes were calculated using pooled SD data and were classified as trivial (<0.20–0.49), moderate (0.50–0.79) and large (>0.80) (Cohen, 1992). All statistical analyses were conducted using PASW Statistics Editor 24.0 for Mac (SPSS, Inc, Chicago, IL, USA), with statistical significance set at $p \leq 0.05$. All data are reported as mean ± SD, unless otherwise stated. Except for the PL data (calculated from an average of the first 15-min measures), between session reliability was assessed using intraclass correlation coefficients (ICC) from the pre-HT measures. The ICC values were then used to calculate the minimal detectable difference (MDD) via the calculation of the standard error of measurement (SEM). SEM was calculated using the formula: $SDPooled \times \sqrt{(1 - ICC)}$ (Thomas, Nelson, & Silverman, 2005), whilst MDD was calculated using the formula: $MDD = SEM \times 1.96 \times \sqrt{2}$ (Weir, 2005).

Results

Sprint times

The GLM identified no significant trial*time interaction for 5 m ($p = 0.299$; $\eta^2 = 0.120$) or 10 m sprint performance ($p = 0.293$; $\eta^2 = 0.126$); however large η^2 values were observed. As illustrated in Table 1, moderate *d* values were observed at 55 min into the protocol, with faster 10 and 5 m sprint times recorded in the RWU trial. Likewise, a similar observation was identified at 60 min with faster 10 m sprint times in the RWU trial. In support of the observed improvements in 5 and

Table 1. Sprint time (s) data recorded during both experimental trials.

	5m		10m		20m	
	CON	RWU	CON	RWU	CON	RWU
0 min	1.18 ± 0.13 (CI = 1.08 to 1.27)	1.13 ± 0.08 (CI = 1.07 to 1.19; *d*= 0.4)	1.98 ± 0.16 (CI = 1.87 to 2.09)	1.93 ± 0.09 (CI = 1.86 to 2.00; *d*= 0.4)	3.63 ± 0.23 (CI = 3.19 to 3.53)	3.34 ± 0.11 (CI = 3.26 to 3.42; *d*= 0.1)
5 min	1.15 ± 0.12 (CI = 1.06 to 1.23)	1.16 ± 0.09 (CI = 1.10 to 1.22; *d*= 0.1)	1.96 ± 0.16 (CI = 1.84 to 2.07)	1.95 ± 0.11 (CI = 1.87 to 2.03; *d*= 0.1)	3.34 ± 0.23 (CI = 3.17 to 3.50)	3.34 ± 0.11 (CI = 3.26 to 3.48; *d*= 0.2)
10 min	1.14 ± 0.10 (CI = 1.07 to 1.21)	1.15 ± 0.10 (CI = 1.08 to 1.22; *d*= 0.1)	1.94 ± 0.11 (CI = 1.86 to 2.02)	1.95 ± 0.08 (CI = 1.89 to 2.01; *d*= 0.1)	3.35 ± 0.16 (CI = 3.23 to 3.46)	3.34 ± 0.16 (CI = 3.25 to 3.47; *d*= 0.1)
15 min	1.13 ± 0.11 (CI = 1.05 to 1.25)	1.14 ± 0.10 (CI = 1.07 to 1.21; *d*= 0.1)	1.93 ± 0.14 (CI = 1.83 to 2.03)	1.94 ± 0.11 (CI = 1.85 to 2.02; *d*= 0.0)	3.34 ± 0.21 (CI = 3.19 to 3.49)	3.34 ± 0.16 (CI = 3.23 to 3.45; *d*= 0.0)
Pre HT	1.12 ± 0.11 (CI = 1.01 to 1.24)	1.12 ± 0.11 (CI = 1.04 to 1.20; *d*= 0.0)	1.93 ± 0.18 (CI = 1.80 to 2.06)	1.80 ± 2.06 (CI = 1.84 to 2.01; *d*= 0.1)	3.35 ± 0.21 (CI = 3.19 to 3.51)	3.33 ± 0.18 (CI = 3.21 to 3.46; *d*= 0.1)
Post HT	1.15 ± 0.14 (CI = 1.05 to 1.25)	1.13 ± 0.09 (CI = 1.06 to 1.91; *d*= 0.2)	1.99 ± 0.14 (CI = 1.89 to 2.10)	1.94 ± 0.09 (CI = 1.88 to 2.00; *d*= 0.4)	3.43 ± 0.19 (CI = 3.29 to 3.56)	3.36 ± 0.18 (CI = 3.23 to 3.48; *d*= 0.4)
50 min	1.16 ± 0.14 (CI = 1.06 to 1.25)	1.12 ± 0.10 (CI = 1.05 to 1.19; *d*= 0.3)	1.97 ± 0.15 (CI = 1.86 to 2.07)	1.93 ± 0.09 (CI = 1.86 to 2.00; *d*= 0.3)	3.39 ± 0.21 (CI = 3.25 to 3.54)	3.33 ± 0.15 (CI = 3.22 to 3.44; *d*= 0.4)
55min	1.15 ± 0.08 (CI = 1.09 to 1.21)	1.09 ± 0.09 (CI = 1.03 to 1.16; d = 0.6)	1.97 ± 0.11 (CI = 1.89 to 2.04)	1.92 ± 0.08 (CI = 1.86 to 1.97; *d*= 0.5)	3.42 ± 0.20 (CI = 3.28 to 3.56)	3.32 ± 0.12* (CI = 3.23 to 3.40; *d*= 0.6; CI *diff*= −0.17 to −0.03)
60 min	1.15 ± 0.11 (CI = 1.08 to 1.23)	1.12 ± 0.07 (CI = 1.07 to 1.17; *d*= 0.2)	1.99 ± 0.11 (CI = 1.91 to 2.06)	1.92 ± 0.06 (CI = 1.88 to 1.97; *d*= 0.7)	3.43 ± 0.17 (CI = 3.31 to 3.55)	3.39 ± 0.17 (CI = 3.27 to 3.50; *d*= 0.7)
ICC	0.889 (CI = 0.973 to 0.537)		0.924 (CI = 0.981 to 0.691)		0.899 (CI = 0.975 to 0.595)	
SEM	0.04 (range = 0.02 to 0.09)		0.04 (range = 0.02 to 0.08)		0.06 (range = 0.03 to 0.12)	
MDD	0.12 (range = 0.06 to 0.25)		0.12 (range = 0.06 to 0.23)		0.17 (range = 0.08 to 0.34)	

Abbreviations: min = minute; CON = control trial; RWU = rewarm up trial; CI = 95% confidence interval; CI *diff*= 95% confidence interval for the difference; ICC = Intraclass correlation coefficient; SEM = Standard error of measurement; MDD = Minimal detectable difference; range = range of measure based on upper and lower CI values for ICC

10 m sprint performance observed at 55 min into the protocol, a significant trial*time interaction for 20 m sprint performance ($p = 0.042$; $\eta^2 = 0.192$) was also identified. The aforementioned differences were all within the range associated with the MDD. The GLM did not, however, identify a significant main effect for trial (5 m: $p = 0.416$, $\eta^2 = 0.075$; 10 m: $p = 0.223$, $\eta^2 = 0.160$; 20 m: $p = 0.252$, $\eta^2 = 0.143$) nor time (5 m: $p = 0.751$, $\eta^2 = 0.065$; 10 m: $p = 0.741$, $\eta^2 = 0.066$; 20 m: $p = 0.086$, $\eta^2 = 0.169$).

Jump height

As identified in Table 2, the GLM identified a significant trial*time interaction for SJ performance ($p = 0.034$; $\eta^2 = 0.289$), with significantly higher values recorded at 5 min and post-HT in the RWU condition, with moderate *d* values also being identified. These observed differences were however less than the MDD and associated range at 5 min but were within the range associated with the MDD post-HT. Moderate *d* values were also recorded at 55 min with lower values recorded in the RWU trial when compared to the corresponding measurement point in the other trial; however, again this difference was less than both the MDD and associated range. The GLM did not, however, identify a significant main effect for trial ($p = 0.250$; $\eta^2 = 0.144$) or time ($p = 0.320$; $\eta^2 = 0.116$).

As identified in Table 2, the GLM identified a significant trial*-time interaction for CMJ performance ($p = 0.024$; $\eta 2 = 0.280$), with data recorded Post HT in the RWU trial being significantly higher than the CON trial. This difference elicited a moderate Cohen's *d* effect size and a difference greater than the MDD. The GLM also identified a significant main effect for trial ($p = 0.040$; $\eta^2 = 0.389$), but not for time ($p = 0.053$; $\eta^2 = 0.226$).

Heart rate

As identified in Table 3, the GLM identified a significant trial*-time interaction for average HR ($p < 0.001$; $\eta^2 = 0.824$), with data recorded at Post HT in the RWU trial being significantly higher than the corresponding time point in the CON trial (*d* =5.8). When comparing between trials, there were some moderate and large *d* values observed pre-intervention; however, the differences were lower than the MDD and associated range. The GLM did not identify a significant main effect for trial ($p = 0.453$; $\eta^2 = 0.064$); however, as identified in Table 3, there was a significant main effect for time ($p < 0.001$; $\eta^2 = 0.581$).

Perceived muscle soreness

As identified in Table 3, the GLM identified no significant trial*-time interaction ($p = 0.668$; $\eta 2 = 0.075$) recorded between trials; however, at post HT, a moderate *d* value was observed, with higher values recorded post-HT in the RWU trial. This difference was however less than the MDD and associated range. The GLM did not, however, identify a main effect for trial ($p = 0.191$; $\eta^2 = 0.182$), but there was a significant main effect for time ($p = < 0.01$; $\eta^2 = 0.688$) with the MS data increasing across each trial.

Table 2. Jump heights (cm) data recorded during both experimental trials.

	SJ		CMJ	
	CON	RWU	CON	RWU
0 min	31.01 ± 6.89 (CI = 26.08 to 35.94)	30.19 ± 3.43 (CI = 27.73 to 32.64; *d*= 0.2)	30.66 ± 5.33 (CI = 26.85 to 34.47)	31.96 ± 4.99 (CI = 28.39 to 35.52; *d*= 0.3)
5 min	29.13 ± 5.53 (CI = 25.18 to 33.09)	31.68 ± 5.05* (CI = 28.07 to 35.29; *d*= 0.5; CI *diff*= 0.73 to 4.36)	30.34 ± 4.26 (CI = 27.29 to 33.38)	31.26 ± 5.22 (CI = 27.53 to 34.99; *d*= 0.2)
10 min	29.37 ± 5.06 (CI = 25.75 to 33.00)	30.17 ± 5.09 (CI = 26.52 to 33.81; *d*= 0.2)	29.35 ± 4.58 (CI = 26.07 to 32.62)	30.20 ± 4.62 (CI = 26.89 to 33.50; *d*= 0.2)
15 min	28.61 ± 5.33 (CI = 24.80 to 32.42)	30.00 ± 6.02 (CI = 25.70 to 34.31; *d*= 0.2)	29.22 ± 5.13 (CI = 25.55 to 32.89)	30.74 ± 5.91 (CI = 26.51 to 34.96; *d*= 0.3)
Pre HT	29.13 ± 5.12 (CI = 25.46 to 32.80)	30.30 ± 4.81 (CI = 26.86 to 33.75; *d*= 0.0)	30.19 ± 4.81 (CI = 26.75 to 33.63)	30.18 ± 5.00 (CI = 26.60 to 33.76; *d*= 0.0)
Post HT	26.96 ± 5.00 (CI = 23.38 to 30.53)	30.17 ± 5.13* (CI = 26.50 to 33.84; *d*= 0.6; CI *diff* =1.16 to 5.27)	28.15 ± 4.72 (CI = 24.77 to 31.52)	31.53 ± 5.43 (CI = 27.65 to 35.43; *d*= 0.7; CI *diff* =2.04 to 4.74)
50 min	26.43 ± 4.62 (CI = 26.12 to 32.74)	30.67 ± 4.57 (CI = 27.40 to 33.95; *d*= 0.3)	29.45 ± 4.50 (CI = 26.23 to 32.68)	30.84 ± 4.57 (CI = 27.57 to 34.11; *d*= 0.3)
55min	32.11 ± 3.05 (CI = 29.92 to 34.29)	29.81 ± 4.08 (CI = 26.88 to 32.73; *d*= 0.6)	31.35 ± 4.45 (CI = 28.17 to 34.54)	31.08 ± 4.38 (CI = 27.94 to 34.21; *d*= 0.1
60 min	30.28 ± 4.96 (CI = 26.73 to 33.83)	30.82 ± 5.30 (CI = 27.03 to 34.61; *d*= 0.1)	30.57 ± 4.44 (CI = 27.40 to 33.75)	31.67 ± 4.34 (CI = 28.57 to 33.75; *d*= 0.3
ICC	0.818 (CI = 0.954 to 0.307)		0.943 (CI = 0.986 to 0.766)	
SEM	2.08 (range = 1.05 to 4.06)		1.14 (range = 0.53 to 2.54)	
MDD	5.76 (range = 2.90 to 11.25)		3.16 (range = 1.47 to 7.04)	

Abbreviations: min = minute; CON = control trial; RWU = rewarm up trial; CI = 95% confidence interval; CI *diff*= 95% confidence interval for the difference; ICC = Intraclass correlation coefficient; SEM = Standard error of measurement; MDD = Minimal detectable difference; range = range of measure based on upper and lower CI values for ICC

Rating of perceived exertion

As identified in Table 3, the GLM identified a significant trial*time interaction for RPE ($p = 0.035$; $\eta^2 = 0.301$), with data recorded at 50 min, 55 min, and 60 min in the RWU trial being significantly higher than the corresponding time points in the CON trial. These differences also elicited moderate and large *d* values, with the differences also being greater than the MDD. The GLM did not identify a significant main effect for trial ($p = 0.105$; $\eta^2 = 0.265$); however, there was a significant main effect for time ($p < 0.001$; $\eta^2 = 0.786$), with values increasing across measurement points.

Table 3. Heart rate, MS, and RPE data recorded during both experimental trials.

	HR ($b \cdot min^{-1}$)		MS (a.u)		RPE (a.u)	
	CON	RWU	CON	RWU	CON	RWU
0 mins	147 ± 12 (CI = 139 to 156)	137 ± 14 (CI = 127 to 147; *d*= 0.8)	11 ± 11 (CI = 5 to 11)	9 ± 6 (CI = 5 to 14; d = 0.2)	10 ± 1 (CI = 8 to 11)	9 ± 1 (CI = 8 to 10; *d*= 0.1)
5 mins	147 ± 11 (CI = 139 to 155)	147 ± 17 (CI = 135 to 159; d = 0.0)	16 ± 12 (CI = 9 to 17)	16 ± 9 (CI = 9 to 22; d = 0.0)	11 ± 1 (CI = 10 to 12)	11 ± 1 (CI = 10 to 12; *d*= 0.3)
10 mins	149 ± 12 (CI = 141 to 168)	145 ± 20 (CI = 130 to 159; d = 0.3)	20 ± 17 (CI = 9 to 25)	20 ± 11 (CI = 11 to 28; d = 0.1)	12 ± 1 (CI = 11 to 12)	12 ± 2 (CI = 10 to 13; *d*= 0.1)
15 mins	159 ± 13 (CI = 149 to 168)	150 ± 17 (CI = 138 to 162; d = 0.6)	23 ± 16 (CI = 13 to 28)	25 ± 18 (CI = 13 to 38; d = 0.1)	12 ± 1 (CI = 12 to 13)	12 ± 1 (CI = 11 to 13; *d*= 0.0)
Pre-HT	151 ± 19 (CI = 138 to 165)	152 ± 17 (CI = 140 to 164; d = 0.1)	36 ± 25 (CI = 18 to 48)	36 ± 24 (CI = 19 to 53; d = 0.0)	13 ± 1 (CI = 12 to 14)	13 ± 1 (CI = 12 to 14; *d*= 0.1)
Post-HT	93 ± 9 (CI = 87 to 99)	160 ± 14* (CI = 150 to 169; d = 5.8; CI *diff*= 57 to 77)	33 ± 19 (CI = 17 to 43)	40 ± 21 (CI = 25 to 55; d = 0.4)		
50 mins	145 ± 18 (CI = 132 to 157)	144 ± 19 (CI = 130 to 158; d = 0.0)	36 ± 18 (CI = 22 to 45)	38 ± 26 (CI = 19 to 56; d = 0.1)	11 ± 1 (CI = 11 to 12)	13 ± 2* (CI = 12 to 14; d = 1.1; CI *diff*= 1 to 3)
55 mins	147 ± 16 (CI = 135 to 158)	141 ± 22 (CI = 126 to 157; d = 0.3)	43 ± 18 (CI = 28 to 52)	42 ± 28 (CI = 23 to 62; d = 0.0)	12 ± 1 (CI = 11 to 13)	14 ± 2* (CI = 12 to 15; *d*= 0.7; CI *diff*= 0 to 2)
60 mins	153 ± 16 (CI = 142 to 165)	145 ± 22 (CI = 129 to 160; d = 0.4)	49 ± 23 (CI = 30 to 61)	51 ± 29 (CI = 30 to 71; d = 0.1)	13 ± 2 (CI = 12 to 14)	14 ± 2* (CI = 13 to 16; *d*= 0.6; CI *diff*= 0 to 2)
ICC	0.756 (CI = 0.941 to 0.063)		0.921 (CI = 0.980 to 0.674)		0.893 (CI = 0.974 to 0.566)	
SEM	9 (range = 4 to 17)		7 (range = 3 to 14)		0 (range = 0 to 1)	
MDD	24 (range = 12 to 47)		18 (range = 9 to 37)		1 (range = 1 to 2)	

Abbreviations: min = minute; CON = control trial; RWU = rewarm up trial; CI = 95% confidence interval; CI *diff*= 95% confidence interval for the difference; ICC = Intraclass correlation coefficient; SEM = Standard error of measurement; MDD = Minimal detectable difference; range = range of measure based on upper and lower CI values for ICC

Table 4. PlayerLoad™ data recorded during both experimental trials.

	PL_V		PL_{AP}		PL_{ML}		PL_{Total}	
	CON	RWU	CON	RWU	CON	RWU	CON	RWU
0-5 mins	32.76 ± 2.13	33.06 ± 4.59	18.57 ± 3.10	18.57 ± 4.06	15.55 ± 2.96	16.43 ± 3.52	66.88 ± 6.48	68.07 ± 8.97
	(CI = 30.79 to 34.73)	(CI = 28.82 to 37.31; d = 0.1)	(CI = 15.71 to 21.43)	(CI = 14.82 to 22.23; d = 0.0)	(CI = 12.81 to 18.29)	(CI = 13.18 to 19.69; d = 0.3)	(CI = 60.88 to 72.87)	(CI = 59.77 to 76.37; d = 0.2)
5-10 mins	31.89 ± 2.54	32.50 ± 4.20	18.25 ± 3.37	18.10 ± 4.31	15.12 ± 3.41	15.85 ± 2.91	65.26 ± 7.77	66.45 ± 8.40
	(CI = 29.54 to 34.23)	(CI = 28.61 to 36.39; d = 0.2)	(CI = 15.14 to 21.37)	(CI = 14.12 to 22.09; d = 0.0)	(CI = 11.97 to 18.28)	(CI = 13.16 to 18.54; d = 0.2)	(CI = 58.08 to 72.45)	(CI = 58.68 to 74.22; d = 0.1)
10-15 mins	33.80 ± 2.92	34.62 ± 4.82	19.20 ± 3.87	18.79 ± 4.02	15.88 ± 3.54	16.73 ± 3.88	68.88 ± 8.08	70.14 ± 9.54
	(CI = 31.10 to 36.50)	(CI = 30.16 to 36.39; d = 0.1)	(CI = 15.61 to 22.78)	(CI = 15.07 to 22.51; d = 0.1)	(CI = 12.61 to 19.16)	(CI = 13.14 to 20.32; d = 0.2)	(CI = 61.41 to 79.36)	(CI = 61.31 to 78.96; d = 0.1)
45-50 mins	33.20 ± 3.21	33.94 ± 4.26	18.58 ± 3.84	17.85 ± 4.34	15.16 ± 2.92	15.43 ± 3.73	66.94 ± 8.18	67.22 ± 9.94
	(CI = 30.24 to 36.17)	(CI = 30.00 to 37.87; d = 0.2)	(CI = 15.02 to 22.13)	(CI = 13.84 to 21.86; d = 0.2)	(CI = 12.46 to 17.86)	(CI = 11.98 to 18.88; d = 0.1)	(CI = 59.38 to 74.50)	(CI = 58.02 to 76.41; d = 0.0)
50-55 mins	34.01 ± 3.03	34.36 ± 5.23	19.16 ± 3.97	17.65 ± 4.37	15.26 ± 2.87	15.16 ± 3.77	68.42 ± 8.08	67.17 ± 10.98
	(CI = 31.20 to 36.82)	(CI = 29.52 to 39.21; d = 0.1)	(CI = 15.49 to 22.83)	(CI = 13.61 to 21.69; d = 0.4)	(CI = 12.60 to 17.91)	(CI = 11.67 to 18.64; d = 0.0)	(CI = 60.95 to 75.89)	(CI = 57.01 to 77.33; d = 0.1)
55-60 mins	34.01 ± 3.24	34.24 ± 4.67	19.26 ± 4.26	17.79 ± 4.58	15.44 ± 2.87	15.64 ± 3.37	68.71 ± 8.61	67.67 ± 9.93
	(CI = 31.02 to 37.01)	(CI = 29.93 to 38.56; d = 0.1)	(CI = 15.32 to 23.20)	(CI = 13.55 to 22.03; d = 0.3)	(CI = 12.79 to 18.10)	(CI = 12.52 to 18.75; d = 0.1)	(CI = 60.75 to 76.68)	(CI = 58.49 to 78.86; d = 0.1)
ICC	0.815 (CI = 0.968 to 0.132)		0.803 (CI = 0.967 to 0.344)		0.906 (CI = 0.984 to 0.527)		0.944 (CI = 0.990 to 0.709)	
SEM	1.49 (CI = 0.62 to 3.22)		1.60 (CI = 0.66 to 2.93)		0.99 (CI = 0.41 to 2.23)		1.83 (CI = 0.78 to 4.19)	
MDD	4.13 (CI = 1.71 to 8.94)		4.45 (CI = 1.82 to 8.11)		2.75 (CI = 1.13 to 6.17)		5.09 (CI = 2.15 to 11.60)	

Abbreviations: min = minute; CON = control trial; RWU = rewarm up trial; CI = 95% confidence interval; CI *diff*= 95% confidence interval for the difference; ICC = Intraclass correlation coefficient; SEM = Standard error of measurement; MDD = Minimal detectable difference; range = range of measure based on upper and lower CI values for ICC

Playerload^{tm} metrics

As identified in Table 4, the GLM identified no significant trial*-time interaction for PL_{AP} ($p = 0.199$; $\eta^2 = 0.207$), PL_{ML} ($p = 0.210$; $\eta^2 = 0.222$), PL_V ($p = 0.976$; $\eta^2 = 0.026$), and PL_{Total} ($p = 0.497$; $\eta^2 = 0.130$). There was also no significant main effect for trial observed for PL_{AP} ($p = 0.477$; $\eta^2 = 0.088$), PL_{ML} ($p = 0.526$; $\eta^2 = 0.070$), PL_V ($p = 0.615$; $\eta^2 = 0.045$), and PL_{Total} ($p = 0.811$; $\eta^2 = 0.010$). With the exception of the PL_V ($p = 0.015$; $\eta^2 = 0.361$) data, there was also no significant main effect for time observed for the other PL metrics (PL_{AP}: $p = 0.312$; $\eta^2 = 0.172$; PL_{ML}: $p = 0.177$; $\eta^2 = 0.264$; PL_{Total}: $p = 0.171$; $\eta^2 = 0.218$). For all PL metrics, small *d* values were observed at all between trial measurement points.

Discussion

The aim of the current study was to investigate the influence of a 3-min HT RWU strategy, when compared to a passive HT period in relation to both physical performance and the perceptual and physical response to soccer-specific activity. In support of the study hypotheses, the HT RWU strategy did elicit significant improvements in 20 m sprint, SJ height, and CMJ height performance. Improvements in both 5 m and 10 m sprint performance were also observed following the HT RWU, with moderate effect sizes being identified. Both HR and RPE ratings were found to be significantly higher immediately following the completion of the HT RWU protocol, with the RPE measures also being remaining significantly elevated throughout the first 15-min period of the second half. There were however, no significant differences in MS ratings and PlayerLoad^{TM} following the HT RWU.

The current study was novel in its design by including performance, physical, and perceptual response measures at 5-min periods across the first 15-min period of each half. The use of a SSEP allows for additional measures to be recorded whilst maintaining methodological rigour. A previous study by Lovell et al. (2013b) utilised the same SSEP to assess the effectiveness of different HT RWU strategies; however, the authors only considered performance effects over 15-min epochs, thus potentially missing performance benefits that occur over shorter periods. In contrast, with the study by Lovell et al. (2013b), the current study identified improvements in sprint performance following the HT RWU. Significantly faster 20 m sprint times were identified at 55 min, likewise, although not statistically significant, moderate *d* values were observed for both 5 and 10 m sprint time at 55 min and for 10 m and 20 m sprint times at 60 min. These data, therefore, suggest that the HT RWU strategy utilised in the current study was therefore beneficial for sprint performance; however, these differences were not evident immediately following the RWU and, with the exception of the 10 m sprint times, were not evident 15-min post HT. Acknowledging differences in methods between studies, the current data is in support of previous studies which have identified improved sprint performance following a HT re-warm-up intervention (Edholm et al., 2015; Mohr et al., 2004; Towlson et al., 2013; Yanaoka et al., 2018). The lack of observed improvement in sprint performance at 50 min may be due to the mechanisms associated with the observed performance benefit at 55 and 60 min. For example, it has previously been suggested that a prior bout of high-intensity activity and plyometric type movements may elicit a post-activation potentiation (PAP) effect on sprint performance; however, this response takes ~7 min to materialise (Healey & Comyns, 2017). This could, therefore, explain the observed response 10–15-min post RWU intervention.

Sprint performance has previously been acknowledged as having large implications for match-performance, with Faude, Koch, and Meyer (2012) identifying that 45% of goals scored in the German first division were preceded with straight-line sprints. The observed improvements in sprint performance identified in the current study could, therefore, have potential implications for improved performance during actual match-play. Likewise, when considering that the initial stages of the second half of match play have been associated with reductions in locomotive activity (Towlson et al., 2013); the observed improvements in sprint performance could have potential performance benefits. The mechanisms behind the observed responses have not however been considered in the current study. For example, previous studies have examined core muscle temperature as a potential mechanism to explain observed changes in performance (Lovell, Midgley, et al., 2013; Mohr et al., 2004). Likewise, alterations in oxygen uptake, blood flow, and neuromuscular function have also been proposed as other possible mechanisms (Hodgson, Docherty, & Robbins, 2005).

In support of the performance benefit observed in the sprint data, both the SJ and CMJ performance was improved during the Post-HT assessments. Significantly higher jump height was also identified at 5 min during the RWU trial; however, these differences were less than the range associated with the MDD. This observed difference in the first half may be a result of slight differences in pre-trial practices between trials; however, these observed differences in the first half were not evident at HT, with the protocol seemingly reducing performance to the same absolute thresholds. The immediate improvement in jump performance was not however maintained throughout the first 15-min period of the second half, with SJ performance in the RWU trial actually being reduced with moderate effect sizes at 55 min when compared to the control trial. The observed immediate improvements in jump performance may be attributable to the warm-up having a beneficial effect when compared to passive rest; however, the lack of maintenance associated with this performance improvement may be due to the body-weight plyometric elements of the RWU eliciting a limited PAP effect on jump performance (Tillin & Bishop, 2009). Likewise, although not supported by the sprint data, the observed response with the jump data may be due to a potential fatigue response as a result of the RWU.

It has previously been suggested that when compared to bodyweight exercises, heavy resistance exercises may result in an increased PAP effect (Edholm et al., 2015). As such, further improvements in jump performance could potentially be achieved via the manipulation of the current RWU protocol. It should, however, be acknowledged that the current RWU method was developed to incorporate limited equipment, thus increasing potential practical application. The performance level of the current participants may also be a factor,

with previous literature identifying that more elite players possess increase fatigue resistance and the ability to better elicit a PAP effect in performance (Tillin & Bishop, 2009). As such, future research should consider these and other RWU methods with alternative populations. Adaptations of the current protocol to include exercises which provide greater muscular stimulation, or with the addition of resistance, such as, but not limited to, weighted vests or sand bags, could increase the chances of inducing additional performance benefits. However, practitioners should always be conscious of the practical application of any intervention.

In relation to the HR data, not surprisingly, the current study identified significantly higher HR data immediately following the HT RWU. This response was however not apparent during the first 15 min of the second half, with higher HR values actually being recorded in the CON trial at all second-half measurement points, with moderate effect sizes observed at 60 min. These data, therefore, suggest a lower physiological response following the RWU; however, this should be considered with caution when considering large and moderate effect sizes were also observed in the first half between trials. With the exception of the post-HT measures all other differences were lower than the MDD and associated range. In support of the elevated HR response observed following the HT RWU, the current study identified an elevated RPE response across the first 15 min of the second half of the RWU trial. In support of previous literature (Yanaoka et al., 2018), these data, therefore, suggest that even with improvements in selected performance measures, the perceived exertion of the players is increased, thus suggesting an offset in the perceived fatigue and preparedness at the beginning of the second half. This response could, in turn, have a negative influence on performance through potential pacing strategies and subconscious downregulation (Noakes, Gibson, & Lambert, 2004; Sampson, Fullagar, & Gabbett, 2015; St Clair et al., 2006). These data also suggest that the observed performance benefits may not be a result of the player's perceptual benefit of the HT RWU. It could, however, be suggested that any further improvements or maintenance of performance may have been subsequently hindered by the participants increased perception of effort. The observed increase in RPE may be attributable to the non-familiar nature of the HT RWU strategy.

The PlayerLoad™ data were not significantly different between conditions, thus suggesting that although the total workload is increased with the inclusion of a HT RWU, movement efficiency is not altered following the HT period. A potential factor that has been previously stated as a reason not to implement a RWU, is to avoid unnecessary player fatigue (Towlson et al., 2013). The data provided from this study may help to overcome these perceptions. For example, although the participant's RPE was elevated following the HT RWU, performance, movement efficiency, HR, and perceived muscle soreness were either improved, or were not significantly different in the initial stages of the second half.

Although the authors have tried to conduct a study utilising measures and methods which could easily be utilised in applied practice, a potential limitation of the current study is that the mechanisms behind the observed responses have not been considered. An additional limitation is that the observed response is potentially specific to the current methods utilised in this study. As such, future research may want to consider alternative RWU methods or applying similar methods to other participant groups. Alternative RWU methods may want to consider the use of exercises that promote increase muscular simulation and/or additional resistance to try and develop a further increased performance effect. Any future manipulation of the current RWU method should, however, be completed with the consideration that it needs to be practically relevant and administered in a time frame that has been identified as feasible during a HT period.

Conclusion

The current study identified immediate increases in SJ, CMJ, and 10 and 20 m sprint performance following the completion of a 3-min RWU. Additional performance benefits were also identified for 5, 10, and 20 m sprint performance at 10 min into the second half, and for 10 and 20 m sprint performance 15 min into the second half. These data, therefore, suggest that the current RWU strategy elicit performance benefits during the initial stages of the second half of match-play. The discrepancy in the rate of development and maintenance of these performance responses appears to suggest differences in the mechanisms associated with these observations; however, these mechanisms have not been considered in the current study. Nevertheless, the current study suggests that the current RWU strategy is beneficial for soccer performance, especially when considering the typical reduction in workload following a passive HT period, and the association between successful jumping and sprinting performance for goal-scoring opportunities in soccer.

Disclosure statement

No potential conflict of interest was reported by the authors.

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